Brief Report

This brief summarizes the five turnout outcomes, five variable blocks, and the most important findings from the spatial-error modelling work. The five outcomes are municipal turnout, provincial turnout, federal turnout, the federal-minus-municipal turnout gap, and mean turnout. Across the modelling process, spatial-error models consistently outperformed non-spatial OLS models, so the interpretation should treat turnout as spatially clustered across neighbouring census tracts.

Block	Main variables	Key finding / significant pattern
Block 1: Basic demographics	Age structure, median age, household size, education, low income, unemployment.	Turnout is generally higher in older and more university-educated areas, and lower in younger, larger-household, and higher-unemployment areas. Education is one of the most consistent positive predictors, especially for mean and federal turnout.
Block 2: Housing and residential stability	Renter share, owner share, apartment share, condo share, apartment/condo counts and densities, same-address stability.	Renter share is generally negative, while apartment and condo-related variables often become positive. Zack's point is supported: apartment-heavy areas may reflect owned condo apartments rather than rental apartments. Condo share and apartment density are worth keeping in future models.
Block 3: Immigration, citizenship, and eligibility	Immigrant share, recent immigrant share, non-citizen share, citizen adult share, visible-minority share, language variables.	Visible-minority share, immigrant share, non-citizen share, and non-official mother tongue share are often associated with lower turnout levels, especially municipal and mean turnout. For the federal-minus-municipal gap, some immigration-related variables flip direction, suggesting relatively stronger federal participation than municipal participation in these areas.
Block 4: Election competitiveness and fragmentation	Mayoral margins, vote fragmentation, effective number of candidates/parties.	Greater candidate fragmentation and a higher effective number of mayoral candidates are generally associated with lower turnout. This is somewhat counterintuitive if competition is expected to mobilize voters, and may reflect fragmented political information or less meaningful competitiveness.
Block 5: Municipal-service relevance / local state contact	Park access, transit/no-car measures, social housing, 311 requests, school-age population, road-safety exposure, development applications, shelter/service proximity, polling-place distance.	Park access is one of the strongest positive predictors. Development pressure tends to be negative in combined models, especially for mean/federal turnout. School-age share is often negative in robustness checks. Polling-place distance is negative for municipal and provincial turnout, although the relationship can be nonlinear.

Synthesis for the Empirical Story

The emerging empirical story is that turnout is highest in census tracts that are educated, older, park-accessible, and more connected to condo/apartment ownership patterns. Turnout is lower in places with younger populations, larger households, unemployment, renter concentration, visible-minority/immigrant concentration, and development pressure.

Mean turnout is especially shaped by education, household size, unemployment, visible-minority share, renter share, apartment/condo structure, candidate fragmentation, park access, and development applications. Because spatial error remains important throughout, future modelling and interpretation should continue using spatial-error models rather than plain OLS.